

Attentional Divergence from a Numerosity Discrimination Task Reveals an Overall Change in Boundary Separation in the EZ-Drift Diffusion Model

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INTRODUCTION (Written by Diana and Arina)

Attention is essential in how individuals process visual information and make accurate decisions, particularly when tasks require rapid judgments under uncertain conditions. In everyday environments, attention is often divided across various sensory inputs. This can reduce cognitive processing efficiency, especially in tasks that rely on rapid perceptual discrimination (Schweizer & Koch, 2002). One way to study this is through numerosity discrimination tasks, where individuals must quickly determine which of two quantities is greater based on visual evidence. Performance on these tasks depends not only on the perceptual difficulty of the stimulus but also on the availability of attentional resources.

The Drift Diffusion Model (DDM) provides a useful framework for understanding how perceptual decisions are made by describing how evidence accumulates over time until a response is made. Instead of assuming decisions as immediate, the model proposes that individuals continuously process noisy sensory information until there is sufficient evidence to reach a decision threshold. Within this framework, differences in performance are explained through cognitive parameters. These cognitive parameters reflect internal mental processes involved in decision making such as the speed of evidence accumulated, efficiency, and amount of information required before a response occurs (Myers et al., 2022).

According to the DDM, faster and more accurate decisions occur when evidence is accumulated efficiently, thus allowing the decision boundary to be reached quicker and with less uncertainty. On the other hand, when attention is limited by distractions, it may slow the accumulation and increase error rates (Myers et al., 2022). It demonstrates why DDM is a useful tool for examining how attentional restrictions impact perceptual decision making, especially in tasks that require quick judgement.

Research on divided attention suggests that cognitive resources are limited and must be shared between tasks. When individuals are required to perform multiple tasks simultaneously, performance often declines due to this limited capacity. For example, studies using a dual-task design have shown that increasing the difficulty of one task leads to decreased accuracy and slower response times in the other task, reflecting a trade-off in attention allocation (Liang et al., 2025). Moreover, there is classic work on dual task interference that demonstrates how humans experience a central processing bottleneck that prevents efficient simultaneous task execution (Pashler, 1994). During both tasks, attention is drawn from a central pool of cognitive resources, forcing individuals to prioritize one task over another. As a result, these findings suggest that divided attention conditions typically produce reduced performance compared to single-task conditions.

One line of research suggests that a single system does not handle numerosity perception but rather relies on multiple mechanisms, depending on the number of items being processed. For example, a study on visual numerosity found evidence for three distinct systems. The first is subitizing or instantly recognizing the number of items which are used for smaller quantities. The second is an approximate number system, or an estimation, used for intermediate values. The last is a texture-density system, where the number of items is highly dense and can't be separated, instead merging into a "texture," and is used for very large quantities (Anobile et al., 2020). Importantly, attention plays a key role in numerosity tasks, as performance for increasing numericities is more disrupted under attentional limitations. Building on this, other research using dot-based numerosity tasks shows that spatial properties of the stimulus, such as spacing and arrangement, can strongly influence how many items are perceived. For instance, studies using random dot patterns have demonstrated that perceived numerosity is not solely based on the actual number of dots but is also shaped by the spatial proximity between them, as patterns with more spread-out elements tend to be judged as more numerous (Allik et al., 1991). This suggests that participants rely on visual features like density or spacing as a proxy for numbers, rather than counting directly. Therefore, variations in color grouping and spacing may influence how efficiently participants deduce numerical information under divided attention.

In addition to visual factors, auditory distractions can also significantly impact cognitive performance. Research has shown that background auditory stimuli, especially meaningful or speech-based sounds, can interfere with attention, working memory, and task performance. Higher-intensity auditory distractions have been found to reduce both reaction time speed and accuracy in cognitive tasks, suggesting that competing sensory inputs place additional demands on limited attentional resources (LaPointe et al., 2007). This aligns with resource allocation models discussed earlier, which propose that attention must be divided among competing stimuli, often leading to decreased efficiency in processing task-relevant information.

The present study examines how divided attention affects performance on a numerosity discrimination task. A comparison is made between a single-task condition, in which participants focus on the visual judgment, and a dual-task condition that introduces auditory and motor distractions. Like prior dual-task studies, participants are required to complete a primary visual task while responding to an additional auditory stimulus, allowing measurement of how attentional resources are distributed across tasks. This design enables an evaluation of how multitasking affects accuracy, reaction time, and cognitive mechanisms that underlie perceptual decision-making.

We predicted that participants in the dual-task condition would show a lower drift rate compared to the single-task condition, reflecting less efficient accumulation of color-ratio evidence when attention is divided. This should result in longer reaction times and lower accuracy in decision-making as participants rely on perceptual features such as spacing and density to estimate quantity, a process that becomes less efficient under attentional constraints. Additionally, auditory distraction has been shown to interfere with cognitive processing by increasing cognitive load, further reducing the resources available for visual decision-making.

METHODS (Written by Louise and Tristan)

The study included 30 undergraduate students at the University of Wisconsin-Madison as participants. We used a within-subjects design, allowing every participant to complete both the single-task and dual-task conditions. Participants completed the single-task session first to establish a baseline, followed by the dual-task session in

order to measure the effects of interference. This allowed us to compare the individual performance changes across the two paradigms.

For the visual task of each paradigm, we used a Numerosity Discrimination Task (NDT) which consisted of 50 trials. Each trial presented a static image containing exactly 80 circles, divided between blue and yellow. The coherence level determined the ratio of blue to yellow and was randomized between 0.05 – 0.35 in order to make the NDT sufficiently difficult while maintaining randomization between trials (Nieder et al., 2009). Participants were instructed to choose if there were more blue or yellow circles as quickly and accurately as they could by pressing the ‘b’ and ‘y’ keys respectively. Participants began by completing the single task which consisted of solely the NDT, and reaction time and accuracy were measured for each trial. This first paradigm acted as a baseline to see how they perform without any distractions.

After completion of the single-task paradigm, participants immediately began the dual-task paradigm which consisted of a distinct NDT simultaneous with an audio task. Reaction time and accuracy were measured for both tasks. The audio task required participants to press the spacebar each time they heard a beep. Furthermore, the task was not temporally localized to each trial of the NDT but rather occurred independently during the entire duration of the dual-task paradigm. This temporally decoupled design was necessary to account for physiological motor delays bleeding into subsequent trials. Because NDT trials occur within a short time window, we found participants were unintentionally pressing the spacebar in subsequent trials, contaminating the audio task data with mostly misses and false positives. Therefore, we coded the audio task to beep randomly every 1.5 – 3.5 seconds, causing the stimulus to be frequent yet unpredictable. This paradigm was intended to create a bottleneck in the brain. According to Pashler (1994), a “bottleneck” occurs when the brain is forced to press two different tasks at the same time. This causes a delay because it cannot handle both simultaneously. Research from Lian et al. (2025) explains that human attention is like a “pie” or a limited pool of mental energy that must be shared. Therefore, this dual-task setting forced participants to divide their limited resources between seeing the circles and hearing the sounds, naturally leading them to slower and less accurate decisions.

To gain insight into the neural implications of the task, we used a Large Language Model (Gemini) to fit our data to the Drift Diffusion Model (DDM) as well as to perform several statistical tests (paired t-test and Pearson product-moment correlation test). The DDM is a framework that explains how people make decisions by “accumulating evidence” over time until they obtain enough information to choose (Ratcliff et al., 2008). We opted to use the EZ-DDM developed by Wagenmakers et al. (2007) which is better suited for our smaller population size (N=30). Additionally, the EZ-DDM excels at comparing differences between groups under different conditions. This simplified turning reaction time and accuracy into cognitive scores while maintaining the integrity of the data.

RESULTS & DISCUSSION (Written by Tristan)

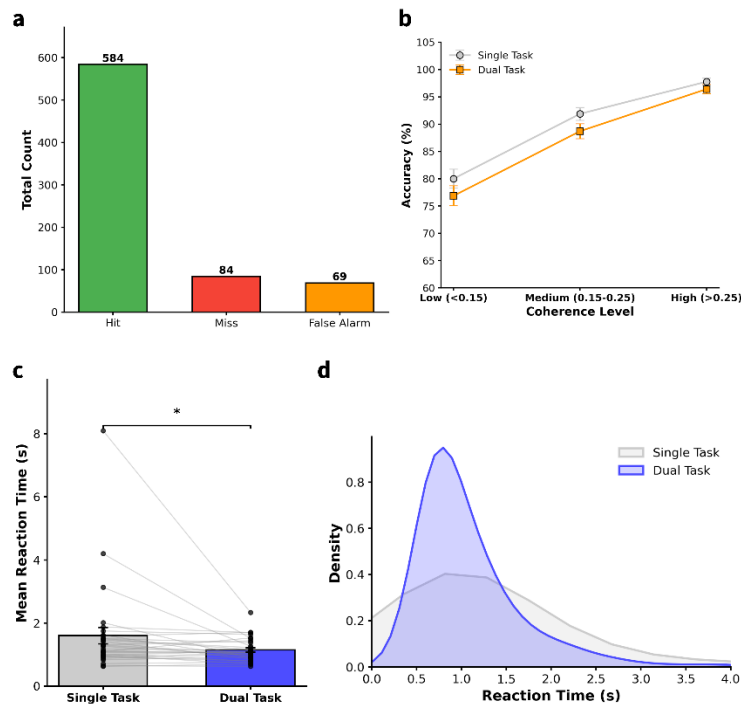


Fig. 1 | Behavioral evidence reveals a speed-accuracy tradeoff. **a**, Total outcomes for the concurrent auditory monitoring task (N=30). **b**, Psychometric functions demonstrating average visual task accuracy distributed across binned coherence levels. **c**, Mean overall accuracy for the visual task was significantly lower during the dual-task paradigm ($t(29) = 2.55$, $p = 0.016$, Cohen's $d = 0.47$, 95% CI = [-4.69%, -0.51%]). **d**, Reaction time density distributions revealing significantly faster responses under dual-task load compared to the single-task baseline ($t(29) = 2.11$, $p = 0.043$, Cohen's $d = 0.39$, 95% CI = [-0.90 s, -0.01 s]).

Behavioral Evidence of a Speed-Accuracy Tradeoff

Implementation of the audio task effectively diverged attentional resources from the numerosity discrimination task (NDT) within the dual-task paradigm. Participants sufficiently engaged in the audio task (Fig. 1a). The 79.2% hit-rate shows a level of attentional resources were effectively applied. Participant's accuracy in the NDT significantly dropped across all coherence levels ($t(29) = 2.55$, $p = 0.016$, Cohen's $d = 0.47$, 95% CI = [-4.69%, -0.51%]) (Fig. 1b-c). The density plot in Figure 1d helps visualize how

participant's reaction times decreased in the dual-task paradigm despite an increased error rate and impaired attentional resources. This was inconsistent with our hypothesis which predicted increased reaction times due to a lower drift rate. We decided to turn to computational modelling in order to investigate the nature of this result.

Computational Modelling Reveals the Variable of Change

We fit our data to the EZ-Drift Diffusion Model (Fig. 2a) and identified boundary separation (a) rather than drift rate (v) as the variable of change between the two task paradigms. Perceptual efficiency remained unchanged ($t(29) = 0.71$, $p = 0.481$, Cohen's $d = 0.13$, 95% CI of difference = $[-0.010, 0.020]$), ultimately allowing us to reject our hypothesis. Rather, boundary separation saw a highly significant decrease while under the audio task load ($t(29) = 2.93$, $p = 0.007$, Cohen's $d = 0.53$, 95% CI of difference = $[-0.144, -0.025]$), accounting for both the decreased accuracy and decreased reaction time (Fig. 3a). Therefore, we found the speed-accuracy tradeoff is best accounted for by an executive shift in decision thresholds, causing less evidence accumulation before making a decision in order to free up working memory for any other tasks imposed on the user.

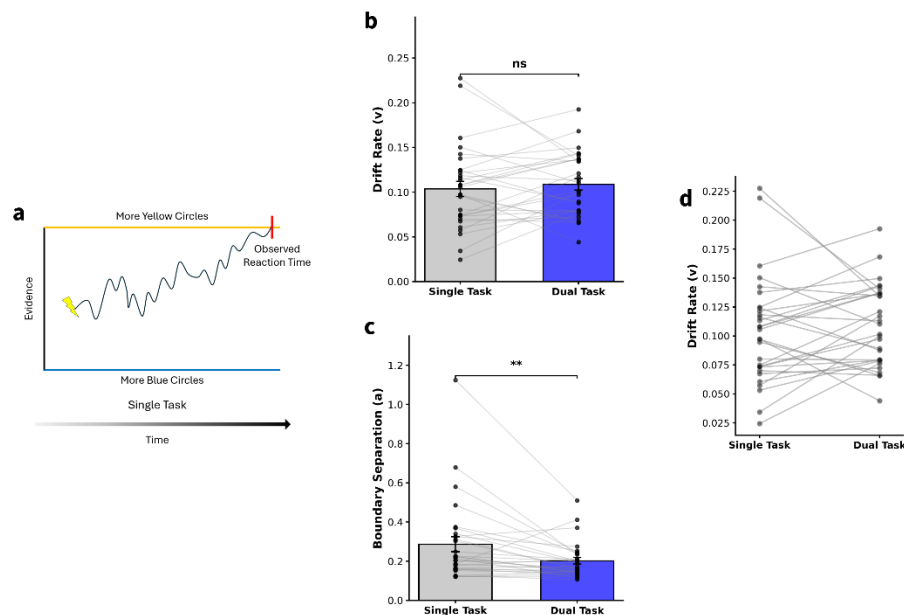


Fig. 2 | Computational modeling reveals change in boundary separation with no change in drift rate. **a**, Example of the DDM. **b**, Computational modeling revealed no significant change in drift rate (v) across the two task paradigms ($t(29) = 0.71$, $p = 0.481$, Cohen's $d = 0.1395$, 95% CI of difference = $[-0.010, 0.020]$), but **c**, showed a highly significant decrease in boundary separation (a) ($t(29) = 2.93$, $p = 0.007$, Cohen's $d = 0.53$, 95% CI of difference = $[-0.144, -0.025]$). **d**, Representative individual slopes of v illustrate a lack of effect on perceptual processing between the two task paradigms.

Executive Strategy Predicts Audio Task Error Rate

Participants who saw a greater change in their decision boundary (Δa) generally had better success during the audio task (Fig. 3b). While our data does not reflect strict significance ($p = 0.121$), the results are consistent with a higher Δa predicting a lower error rate. While requiring more research, we posit a resource bottleneck, where the brain has limited resources and must distribute them across any present tasks in the environment.

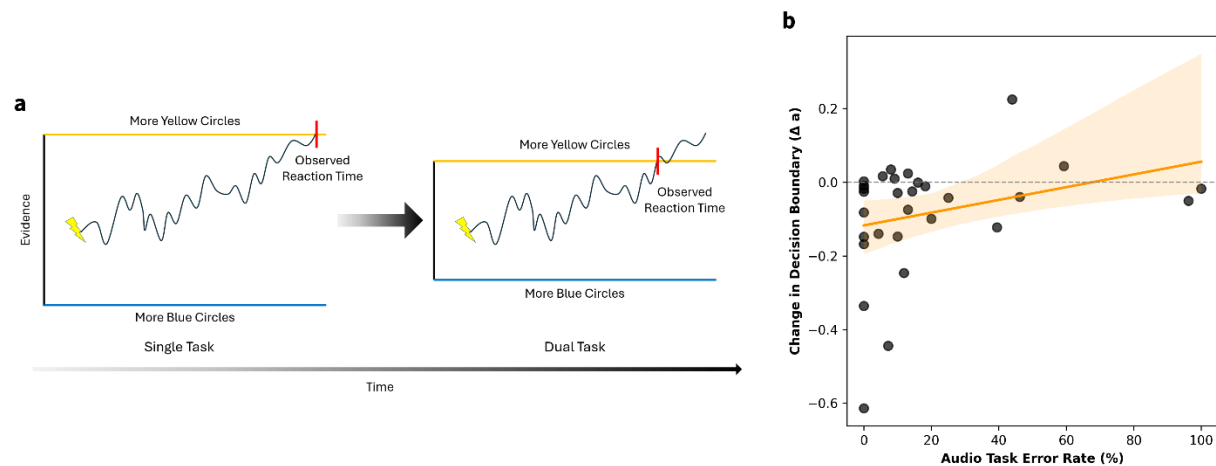


Fig. 3 | Change in decision boundary reflects degree of success in audio task. **a**, Theoretical framework of the DDM reflecting a change in decision boundary (**a**) **b**, Magnitude of Δa suggests a correlation with error rate in the audio task ($r = 0.289$, $p = 0.121$, 95% CI of $r = [-0.079, 0.588]$).

Neurobiological Implications and Limitations

The most notable implication from the data is the highly significant change in boundary separation between the two paradigms. Particularly in recent studies, the subthalamic nucleus (STN) has been shown to play an important role in decision thresholds and overall impulsive behavior. Increases in STN theta activity have been shown to correlate with a positive Δa when accuracy is prioritized over speed (Herz et al., 2024). The STN's involvement in the indirect and hyperdirect pathways (Nambu et al., 2000, Frank et al., 2006) as well as the implications of executive strategy based on the data warrants further research into the relationship between the prefrontal cortex (PFC) and the STN when analyzed under the DDM. Notably, the PFC activates the STN in the hyperdirect pathway, which has downstream effects in keeping the thalamus

inhibited (Frank et al., 2006). Lack of STN activation by a “distracted” PFC could in turn reflect a negative Δa (Fig. 3a).

Another important result was the lack of change in drift rate between the two paradigms. Previous research determined the lateral intraparietal cortex in rhesus monkeys accumulates evidence for or against a particular behavioral response (Roitman et al., 2002). Later research in human patients mapped EEG signals to the DDM and found evidence accumulation occurred in the parietal lobe (Kelly et al., 2013). Our data shows the rate of evidence accumulation (v) remains unchanged. Therefore, future studies measuring a lack of change in the functioning of these brain areas under a similar task structure would provide more insight into the neural correlates of drift rate.

Future studies should use neuroimaging techniques to further investigate the change in boundary separation and the lack of change in drift rate observed in the dual-task paradigm. Our analysis used the EZ-DDM which has its advantages in detecting certain experimental effects (van Ravenzwaaij et al., 2017), but future studies could take advantage of the more robust analysis the DDM provides. Additionally, a narrow range of coherence levels (0.05 – 0.35) were used in order to make the NDT sufficiently difficult. Participants were more successful than anticipated at even these lower coherence levels. The range we used corresponds to a portion of a full psychometric curve (Fig. 3b), but a full curve could be visualized with a full range of coherence levels and a larger population size. Finally, our population consisted of mostly college students which could introduce some confounds, especially in results involving the PFC.

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